



The Ratio

Artist's Proof 03

Relativity

c derived as the conjugacy of propagation and resistance

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Artist's Note

This is Artist's Proof 03 of The 420 Code. It opens Notebook II — Spacetime. The paper asks a question prior to physics: how do the fundamental constants that define the arena arise?

Specifically, are c and G independent constants, or are they conjugate expressions of a single symmetry-breaking event?

Eight sections carry the work. Section 1 names the two speed limits and proposes they are siblings. Section 2 shows that at the moment of symmetry breaking, only ε is available as a scale, and both c and G must inherit from it. Section 3 proves the minimal break produces exactly two sectors via the $\mathbb{Z}_2$ representation theorem. Section 4 derives c from a minimal pre-geometric action and lands the conjugacy of c and G . Section 5 identifies the cosmological origin and the black hole event horizon as the same marginal surface $\varepsilon = 0$, traversed in opposite directions. Section 6 connects the small ε to the observed dark-to-visible energy ratio. Sections 7 and 8 carry the formal mathematics: the action principle and the $\mathbb{Z}_2$ representation theorem.

AP03 is logically independent of AP01 and AP40. Its two postulates (E0, E1) are self-contained. Its derivations require only standard representation theory and scalar field theory. The paper can be read on its own; within The 420 Code, AP20 adds the structural upgrade that turns the conjugacy of c and G from conjecture to theorem.

A note on status. AP03 was originally published as a conjecture: c was derived from the pre-geometric action; G was conjectured to be determined by ε but not derived. The central open problem was the derivation of G . That problem is structurally resolved. AP20's forcing proof — Axiom B forces exactly one parameter — makes the conjecture E-G a theorem consequence: G cannot be independent of ε . The explicit scaling $G = \kappa/(\varepsilon\Lambda^2)$, equivalently $c/G \propto \varepsilon$ (correcting AP03's original $c/G \propto \varepsilon^2$), was derived in subsequent work from the structural condensate density of section 7 and verified numerically. The conditional claims (E4, E-Identity, E-Hawking) inherit the upgrade. A note on level: AP03's $G = \kappa/(\varepsilon\Lambda^2)$ is a substrate-level scaling — it fixes how G responds to ε with the loop coefficient κ and the cutoff Λ free, anchored at order of magnitude (the dimensionless hierarchy $\alpha/\alpha_{\text{G}}(e) \approx 10^{43}$). It is superseded as the statement of G 's value by AP28 (Notebook IV), which derives $G = \alpha_{\text{em}}^{21}(1 + 1/\pi)\hbar c/m_e^2$ parameter-free ($G \propto \varepsilon^{21}$, 0.69% from measurement) and recovers

the same hierarchy as $\alpha_{\text{em}}^{-20}/(1 + 1/\pi)$. The two operate at different architectural levels (substrate condensate vs. Mode-0 channel count) and are reconciled, not in conflict. $G \propto \varepsilon^{21}$ is canonical for the value; AP03's ε^{-1} scaling is retained as the substrate-level derivation AP28's count refines.

Six kill switches are published with this paper. Every load-bearing claim attaches to a structural condition under which the claim would fail. The 420 Code is published copyleft. Free forever. No paywall. No gatekeepers.

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Orientation

Where this paper fits in The 420 Code

The 420 Code is organised across eight Notebooks. This paper opens Notebook II — Spacetime. The chapter sequence runs AP03 (this paper) → AP05 (The Break) → AP10 (The Dimension) → AP08 (The Identity).

Notebook II derives the structure of space, time, and gravity from the axiom. Notebook I (The Premise) established the axiom, named the four conditions {S, B, R, C}, and proved the Embedding Hypothesis. With EH proven, Notebook II runs unconditionally. The fundamental constants c and G , the metric signature, the dimensional count $N = 3$, and Einstein's field equations all follow from {S, B, R, C} once the conditional has been removed.

AP03 sits first in Notebook II because it does the constants. Without c and G , the arena does not exist. The chapter shows that c and G are not independent inputs but conjugate outputs of the same break — both determined by the single breaking amplitude ε . This sets up the rest of Notebook II: AP05 walks the seven-step chain from the axiom to the obligation and reformulates the cosmological identity as junction conditions, AP10 derives $N = 3$, AP08 derives Einstein's field equations. Each chapter inherits AP03's structural setting that both speed limits come from the same source.

AP03 is logically independent of AP01 and AP40 — its two postulates (E0, E1) are self-contained and its derivations require only standard representation theory and scalar field theory. The paper can be read on its own. Within The 420 Code, AP20 adds the structural upgrade: AP20's forcing proof on Axiom B turns the conjugacy of c and G from conjecture to theorem.

How to read this paper

The reading order is front to back. Section 0 names the dependencies and the scope. Sections 1 through 8 carry the structural argument and the formal mathematics. Section 9 is the falsification register. Section 10 is the synthesis — what this paper closes, what it leaves open, where the philosophical-register companions live, and how The 420 Code reads with AP03 in place.

Two voices alternate as the material requires. Narrative passages — openings, transitions, body anchors, the bootstrap problem, the Planck-units whisper, the Big Bang as door — speak directly. Formal sections — postulates, theorems, the action, kill switches — speak in the precision the structure requires. The gear-changes are intentional.

A note on what this paper does and does not do. AP03 derives c from a pre-geometric action, proves the binary decomposition of the broken phase from the $\mathbb{Z}_2$ representation theorem, identifies the structural identity of the cosmological origin and the black hole event horizon as the same marginal surface $\varepsilon = 0$, and lands a qualitative connection between small ε and the observed dark-to-visible energy ratio. It does not derive G from first principles within its own action — the derivation runs from the structural condensate density of section 7 plus AP20's forcing proof. It does not predict the quantitative dark-sector partition; only the qualitative direction (small $\varepsilon \rightarrow$ most content in Mode 0). The substrate parameters α and β in the kinetic term are introduced but not derived from a more fundamental theory. What the paper does is install the conjugacy of c and G as a structural fact and locate the cosmological and black-hole boundaries on the same marginal surface.

0 — Dependency and Scope

0.1 — What this paper does

AP03 derives the conjugacy of the two fundamental speed limits of physics — c and G — from a single symmetry-breaking event with amplitude ε . Both constants emerge from the same break. Neither is independent. Neither is fundamental on its own.

Three claims, each derived or structurally forced.

First. At the moment of symmetry breaking, only the breaking amplitude ε is available as a scale. No spacetime, no mass-energy, no pre-existing coupling — nothing has emerged yet. Both c and G must therefore inherit from ε . This is a structural argument from the bootstrap problem: you cannot use the rules of the house to explain how the house was built.

Second. The minimal break produces exactly two sectors. The pre-state has $\mathbb{Z}_2$ symmetry — the simplest non-trivial symmetry group. By the representation theory of $\mathbb{Z}_2$ (section 8), the broken phase decomposes into exactly two irreducible sectors: a propagation sector (Mode 1) carrying c , and a resistance sector (Mode 0) carrying G . Not three. Not one. Two.

Third. The cosmological origin and the black hole event horizon are the same marginal surface — the surface where ε locally returns to zero — traversed in opposite temporal directions. The Big Bang is the break opening; the event horizon is the break closing. Hawking radiation is the universe refusing to permit a permanent pocket of restored symmetry within a globally broken phase.

The paper makes one further structural claim, qualitatively. The dark-to-visible energy ratio ($\sim 95:5$) is a consequence of small ε : the smaller the break, the more content remains in Mode 0 (the resistance sector), the less spills into Mode 1 (the propagation sector that carries electromagnetic interaction). The qualitative direction is robust; the quantitative mapping requires a model of the initial fluctuation spectrum that is not derived here.

0.2 — Dependencies

The axioms {S, B, R, C} (formal statement carried at AP20 section 3.3; completeness and minimality obligations fenced at KS-P.1 / KS-P.2). AP03 was originally written before the canonical formalisation; the mapping from AP03's postulates onto the canonical axioms is made explicit in section 0.3 below. Axiom S is Symmetry; AP03's E0 specifies the minimal symmetry. Axiom B is the unique break; AP03's E1 instantiates it as the spontaneous breaking $\mathbb{Z}_2 \rightarrow \text{trivial}$ with amplitude ε . Axiom R is the persistence of the record; the broken vacuum is what persists. Axiom C is Constraint — locality; the bounded propagation it produces is why c is finite, G is positive, and ε is small and bounded.

AP20 (The Proof). Load-bearing for the conjugacy upgrade. AP20's Paper D proves Axiom B forces exactly one free parameter. With this proof in place, the AP03 conjecture E-G (G depends on ε) becomes a theorem consequence: G cannot be independent of ε , because if it were, the axioms would admit at least two free parameters. AP20 takes AP03's central conjecture and makes it structurally necessary.

AP06 (The Leakage Constant). Load-bearing for the numerical identification of ε . AP06 derives the cycle form $1:1 + 1 \times \varepsilon @ AS [+1/137 / -1/137]$ and identifies ε numerically with the fine-structure constant $\alpha \approx 1/137$. The gravitational hierarchy $ke^2/(G \cdot m_e^2) \approx 10^{42}-10^{43}$ verification in section 7.5 uses $\varepsilon \approx 1/137$ from AP06.

AP24 (The Residual). Provides the constraint Λ that pins the dependence of m_p on ε through the relation $m_p \propto \sqrt{\varepsilon} \cdot \sqrt{\Lambda}$. AP03 derives the proportionality $m_p \propto \sqrt{\varepsilon}$; AP24 fixes Λ so the relationship is dimensionally complete.

AP08 (The Identity). Dependent on AP03, not the reverse. AP08 derives Einstein's field equations from the record algebra; the gravitational coupling G that appears in AP08's field equations is the same G that AP03 conjectures (and AP20 forces) to be determined by ε . The disclosed circularity in AP08's $G = 2\kappa/m_e^2$ is structural debt D1 of The 420 Code, substantially reduced by the AP20 forcing proof + the section 7 derivation but not fully closed.

0.3 — Axiom mapping

AP03 was written before the canonical axiom set {S, B, R, C} was formalised. The mapping below shows how AP03's postulates instantiate the four axioms.

Axiom S (Symmetry) ↔ Postulate E0. S is the two-sector base condition — identity under exchange, the precondition for distinction to be meaningful. E0 specifies the minimal symmetry of the pre-state: $\mathbb{Z}_2$, the simplest non-trivial group. The pre-state whose existence Axiom S asserts is what E0 endows with the minimal symmetry capable of nontrivial breaking. AP03 chooses $\mathbb{Z}_2$ by parsimony; the choice is tested via KS-R.2 (binary decomposition) — a richer pre-state symmetry would produce more than two irreducible representations, so any falsification of the binary decomposition is also a falsification of $\mathbb{Z}_2$ minimality.

Axiom B (Exactly one break) ↔ Postulate E1. B requires exactly one irreducible symmetry-breaking event. E1 instantiates this as the spontaneous breaking $\mathbb{Z}_2 \rightarrow \text{trivial}$ with amplitude ε . The minimality of $\mathbb{Z}_2$ ensures the break is irreducible: there is no intermediate subgroup between $\mathbb{Z}_2$ and the trivial group. AP20's forcing proof on Axiom B is what upgrades AP03's E-G conjecture (G depends on ε) to a theorem.

Axiom R (The record persists) ↔ The broken vacuum. R requires that the break leave a persistent record. In AP03, the broken vacuum with propagation speed $c = \sqrt{(\beta/\alpha)}$ and condensate energy density $\rho_0 = \varepsilon^2 \Lambda^4 / (4\lambda)$ is the persistent record. The two sectors (Mode 0, Mode 1) persist because the broken vacuum is stable; stability is what makes the record a record.

Axiom C (Constraint) ↔ Finiteness of c and positivity of G. C requires that all quantities remain bounded. In AP03: c is finite (set by the substrate ratio β/α); $G > 0$ (theorem from E-G: $G = \kappa/(\varepsilon \Lambda^2)$); ε itself is small and bounded.

Note: AP03's postulates are more specific than {S, B, R, C}. They choose a particular symmetry group ($\mathbb{Z}_2$) and a particular mechanism (SSB of a scalar field). The axioms {S, B, R, C} are the abstract structure; AP03 is one specific instantiation.

0.4 — Epistemic status per section

section 1 (The Two Speed Limits). FRAMING. Names the two structural speed limits of physics and proposes they are siblings.

section 2 (Before the Arena). STRUCTURAL ARGUMENT. The bootstrap problem and the forcing of ε as the only available scale at the moment of symmetry breaking. Claim E-Bootstrap.

section 3 (The Minimal Break). DERIVATION. The $\mathbb{Z}_2$ representation theorem applied to the broken phase. Postulate E0; Theorem 3.1 (proof in section 8); Claim E2 with KS-R.2.

section 4 (The Conjugacy of c and G). DERIVATION + STRUCTURAL CRITIQUE. Planck units as evidence of conjugacy; the derivation of c from the pre-geometric action (Result A.1); Theorem E-G ($G = \kappa/(\varepsilon\Lambda^2)$) via AP20 forcing proof + the section 7 derivation; the dimensionless formulation as the correct quantitative target.

section 5 (The Origin and the Horizon). DERIVATION. The structural identity Claim E-Identity: cosmological origin and event horizon as the same marginal surface, traversed in opposite directions.

section 6 (The Dark Sector Connection). QUALITATIVE FRAMING. Conjecture E5 that the dark-to-visible energy ratio is determined by ε , qualitatively consistent but not quantitatively derived.

section 7 (The Action Principle). FORMAL DERIVATION. The minimal $\mathbb{Z}_2$ -symmetric pre-geometric action; Result A.1 (Emergence of c); the form of G 's dependence on ε .

section 8 (The $\mathbb{Z}_2$ Representation Theorem). FORMAL PROOF. Theorem B.1 (the only two irreducible representations of $\mathbb{Z}_2$ are the trivial and the sign representation).

section 9 (Kill Switches). FORMAL FALSIFICATION REGISTER. Six kill switches engaged: KS-R.1 through KS-R.6.

section 10 (What This Establishes and What Remains Open). SYNTHESIS.

0.5 — Kill switch summary

Six kill switches are engaged in this paper. The full register is in section 9.

KS-R.1: Independent variability of c and G (gravitational conjugacy). LIVE — EMPIRICAL.

KS-R.2: Non-binary origin. Universe requires more than two fundamental sectors. LIVE — EMPIRICAL.

KS-R.3: Dark ratio independence (mechanism unrelated to c/G). LIVE — EMPIRICAL.

KS-R.4: Horizon inequivalence (cosmological origin and black hole horizon structurally distinct). LIVE — HARD.

KS-R.5: Planck scale mismatch (Planck onset incompatible with $m_p \propto \sqrt{\epsilon}$). LIVE — EMPIRICAL.

KS-R.6: Dimensionless scaling violation ($\alpha \cdot \alpha_G$ at fixed mass varies with ϵ , or α/α_G hierarchy is not consistent with α^2). LIVE — HARD.

0.6 — Structural debts owed

Two structural debts are named explicitly. These are unresolved derivations tied to specific structural claims, tracked at the registry level.

D1 (registry-level, partially closed by AP08 + this paper). The first-principles derivation of G from the axioms without external scaffolding. AP08 derives Einstein’s field equations from the record algebra and identifies G through The Lock ($G = 2\kappa/m_e^2$), with the disclosed structural circularity that κ is itself defined as $Gm_e^2/2$. AP03 contributes by establishing that G cannot be independent of ϵ (via AP20’s forcing proof) and by deriving the explicit form $G = \kappa/(\epsilon\Lambda^2)$ from the structural condensate density of section 7. The induced-gravity calculation used in section 7.5 is named in the literature as Sakharov-Visser; it leans on the standard QFT cutoff structure. The 420 Code does not import that calculation as its source — the load-bearing inputs are the AP20 forcing proof and the structural condensate density, both internal to The 420 Code; the literature calculation is correspondence, confirming the same form from a different route. The closure is partial. The full first-principles derivation — G computed from $\{S, B, R, C\}$ alone, with the structural coefficient κ derived from the broken vacuum response without any reliance on the standard cutoff machinery — remains open. This debt is registry-level, not AP03-specific; AP03’s contribution to the partial closure is recorded here.

D-AP03.1. The substrate parameters α and β in the kinetic term $\mathcal{L} = \frac{1}{2}\alpha(\partial_\tau\varphi)^2 - \frac{1}{2}\beta(\nabla\varphi)^2 - V(\varphi)$ are introduced but not derived from a more fundamental theory. Their ratio gives $c^2 = \beta/\alpha$, but neither α nor β individually is computed from the axioms. This is the deepest open problem in AP03 — distinct from and prior to the G derivation. A truly pre-geometric formulation would derive the

distinction between temporal and spatial coordinates from the axioms rather than presupposing it in the kinetic term.

0.7 — Items held open without claiming debt status

Two further items are bracketed inside this paper. They are real and live, but they are not structural debts in the same sense as D1 and D-AP03.1.

E5 quantitative. The dark-to-visible energy ratio as a function of ε . The paper establishes the qualitative direction (small $\varepsilon \rightarrow$ most content in Mode 0) but does not derive the quantitative mapping. Three issues need resolution before the connection becomes quantitative: the phenomenological distinction between dark matter (clusters gravitationally) and dark energy (does not) within Mode 0; the epoch-dependence of the 95:5 ratio over cosmological history; and the specific functional form linking ε to the partition. This is open derivation work; the qualitative claim stands.

E-Hawking sufficiency. Hawking radiation is structurally motivated in section 4 as the necessary consequence of imposing a global $\varepsilon > 0$ condition on a locally $\varepsilon \rightarrow 0$ region. The paper does not derive the Hawking spectrum from the AP03 action; it argues that the global condition forces leakage and identifies the spectrum derived elsewhere with that leakage. The empirical test of the connection rides on KS-R.4 (horizon inequivalence): if the cosmological origin and the black-hole horizon are shown to be structurally distinct, the structural account of Hawking radiation falls with it.

0.8 — Structural relationships

AP03 opens Notebook II of The 420 Code. AP05, AP10, and AP08 follow as the remaining chapters of Notebook II — the structure of space and time, the dimensional count, and Einstein’s field equations — all inheriting the structural setting AP03 establishes: c and G as conjugate outputs of a single break.

AP20 (Notebook I) upgrades AP03’s conjecture E-G to a theorem via the forcing proof on Axiom B. With AP20 in place, the conditional claims (E4, E-Identity, E-Hawking) inherit the upgrade. AP03 returns the favour by giving AP20’s forcing proof its first quantitative payoff outside the foundational proof itself.

AP06 (Notebook IV) identifies ε numerically with $\alpha \approx 1/137$. AP24 (Notebook IV) constrains Λ . Together they pin AP03's scaling relations to specific numbers: the gravitational hierarchy $ke^2/(G \cdot m_e^2) \approx 10^{42}-10^{43}$, m_p in terms of ε and Λ , the Planck scale as the scale of the break itself.

AP17 (Notebook VI) relates the 0-pole / 1-pole structure of cosmological geometry to AP03's Mode 0 / Mode 1 sectors. AP18 (Notebook VI) derives the MOND floor $a_0 \approx cH_0$ from the axioms, downstream of the small ε .

1 — The Two Speed Limits

You know one of them.

You have felt it your entire life without naming it.

Light crosses the room before your hand finishes reaching for the switch. Nothing outruns it. Nothing negotiates with it.

That limit is c .

There is a second one. You have felt it too, but it hides inside a different experience. You have never been told its name because physics does not yet agree on what to call it.

1.1 — The propagation limit

Special relativity identifies c as the maximum velocity at which information, energy, or causal influence can propagate through spacetime.

Not a property of light alone — a structural constraint on the geometry of spacetime itself.

The Lorentz group enforces c as the invariant velocity linking space and time coordinates.

1.2 — The gravitational coupling and the rate it sets

General relativity identifies gravity not as a force but as spacetime curvature sourced by mass-energy, with G as the coupling that sets the curvature response.

A point of precision. G is a coupling, not a speed. Its dimensions are $\text{m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2}$, not m/s . When the section calls c and G “the two speed limits,” it is not the claim that G is itself a velocity. It is the claim that G fixes the *scale* of a rate that shows up elsewhere in the architecture.

In certain quantum gravity proposals — the gravitational self-energy decoherence models of Diósi and Penrose, and the rate-limiter hypothesis of AP01 Paper B — gravitational self-energy imposes a bound on how fast quantum

superpositions can resolve. The collapse rate in those models is *proportional to* G times other quantities; G is the coupling that fixes the rate's scale, not the rate itself.

How fast possibility becomes definiteness.

The rate-limiter interpretation of G is itself a candidate (AP01 Paper B), not an established result. AP03's conjugacy argument does not depend on the rate-limiter interpretation — only on G being a structural quantity of the broken phase fixed by ε . The bound, where it exists, scales with the mass and spatial extent of the superposition. It is system-dependent, unlike c which is universal.

1.3 — What c and G share

c and G are both non-negotiable structural quantities of the broken phase. They are not the same kind of quantity: c is a velocity (m/s); G is a coupling ($\text{m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2}$). c is universal; the rate G sets, in the rate-limiter picture, is system-dependent.

What they share is the source.

AP03 proposes that c and G are not two independent inputs of physics. They are two outputs of one break, both fixed by the single breaking amplitude ε . The conjugacy is not that both are speeds. The conjugacy is that the same parameter ε , the same break, determines both.

You have lived inside both your entire life.

One governs how fast the light reaches your eye. The other, through the curvature it sets, governs how the world holds together at the scale you live at — and, in the rate-limiter picture, how fast possibility resolves into fact.

The mathematics of one event produces both.

2 — Before the Arena

Picture the moment of symmetry breaking.

Not the Big Bang — the event prior to it. The transition from the symmetric pre-state to the broken state that becomes our universe.

At that exact moment, you have nothing to work with.

2.1 — The bootstrap problem

No spacetime. The geometric arena has not yet formed. c , as a property of spacetime geometry, cannot yet exist as a constraint.

No mass. Mass-energy has not yet differentiated. G , as a coupling to mass-energy, cannot yet exist as a constraint.

Both c and G are properties of the broken state. They emerge from the break.

They cannot govern the break that produces them.

You cannot use the rules of the house to explain how the house was built. The house did not exist yet.

You cannot invoke the speed limits of the universe to explain how the universe acquired its speed limits.

That is the bootstrap problem.

2.2 — The only available constraint

At the moment of symmetry breaking, the only available parameter is the magnitude of the asymmetry itself.

Call it ε .

The break occurs because the symmetric state S is unstable — an infinitesimal perturbation of amplitude ε grows, separating the broken state from the symmetric state. The rate of that separation cannot exceed ε itself, because ε is the only scale in the problem at that moment.

No external clock. No background metric. No pre-existing coupling constant to set the rate.

Claim E-Bootstrap. The first speed limit in the universe is ε . Both c and G inherit from ε . They are downstream expressions of the original constraint, specialised to different degrees of freedom in the broken phase. Status: structural argument. Forced — by the fact that there was nothing else available.

Scope. “ ε is the speed limit” means ε is the only available scale for the growth rate of the unstable mode in the absence of any pre-existing metric or clock. A dimensional and structural claim, not a dynamical derivation.

You have just watched the first constraint appear. Not decreed. Not chosen. Forced.

Everything that follows inherits from that single scale.

3 — The Minimal Break: Why Exactly Two Sectors

3.1 — The symmetry group of the pre-state

The symmetry of the pre-state S is proposed to be $\mathbb{Z}_2$ — the cyclic group of order 2.

Exactly two elements: the identity and a single involution (a transformation that is its own inverse).

The question is not “what symmetry does the pre-state have?” in the empirical sense.

The question is: what is the minimal symmetry that, when broken, produces a nontrivial universe?

A state with no symmetry (the trivial group) cannot break — it is already maximally asymmetric.

A state with $\mathbb{Z}_2$ symmetry is the simplest state that can undergo nontrivial symmetry breaking.

If the universe arises from the minimal possible break, the symmetry group must be $\mathbb{Z}_2$.

Postulate E0 — $\mathbb{Z}_2$ Minimality. The symmetric pre-state S has $\mathbb{Z}_2$ symmetry. The simplest non-trivial symmetry group. Motivated by parsimony but not derived from first principles. Tested via KS-R.2: any falsification of the binary decomposition is also a falsification of $\mathbb{Z}_2$ minimality.

You might object: why $\mathbb{Z}_2$? Why not something richer?

Because richer is not simpler. And the question was: what is the minimum?

The minimum is two.

The formal proof that $\mathbb{Z}_2$ admits exactly two irreducible representations is in section 8.

3.2 — The $\mathbb{Z}_2$ decomposition

When a $\mathbb{Z}_2$ symmetry breaks spontaneously, the state space decomposes into exactly two sectors.

The even sector (+1 eigenvalue): states invariant under the $\mathbb{Z}_2$ involution.

The odd sector (−1 eigenvalue): states that flip sign under the $\mathbb{Z}_2$ involution.

Theorem 3.1 — $\mathbb{Z}_2$ Decomposition. The irreducible representations of $\mathbb{Z}_2$ are exactly two: the trivial representation and the sign representation. Any representation of $\mathbb{Z}_2$ decomposes into copies of these two. There cannot be a third fundamental sector. Proof: section 8.

You cannot argue with this.

It is not a claim about the universe. It is a theorem about $\mathbb{Z}_2$.

The group has two irreducible representations. Not three. Not one. Two.

The mathematics does not negotiate.

Critical distinction: the theorem guarantees two sectors (fundamental categories), not two particle species.

Each sector may contain many dynamical degrees of freedom — multiple particle species, bound states, domain walls, other structures.

The Ising model has $\mathbb{Z}_2$ symmetry; its broken phase contains magnetisation, spinwave excitations, domain walls, and bound states. All belong to either the even or the odd sector.

The representation theorem constrains the number of fundamental categories, not the richness of physics within each category.

3.3 — Identification of sectors

Scope: the following identification of sectors with physical modes is a modelling assumption motivated by the structure of the broken phase, not a theorem consequence.

The $\mathbb{Z}_2$ decomposition (Theorem 3.1) guarantees two sectors. The assignment of physical roles requires an additional argument: the odd sector (V_-), being maximally displaced by the break, is the natural carrier of propagation — content that separates, moves, carries information. The even sector (V_+), being invariant under the break, is the natural carrier of resistance — content that holds, resists separation.

Structurally natural but not unique; an assumption (E-Assignment) that could in principle be falsified independently of E0.

Mode 1 (Propagation sector). The odd sector. Content maximally displaced by the break. Whatever moves, separates, carries information away from the origin. In the broken phase, its limiting velocity becomes c . The entire Standard Model gauge structure lives within Mode 1.

Mode 0 (Resistance sector). The even sector. Content invariant under the break. Whatever stays, holds, resists separation. In the broken phase, its coupling strength is G . The sector may contain internal structure — potentially including the distinct phenomenologies of dark matter and dark energy.

Claim E2 — Binary Decomposition. The symmetry-breaking event produces exactly two fundamental sectors because the symmetry group of the pre-state is $\mathbb{Z}_2$, whose representation theory admits exactly two irreducible representations. The internal structure within each sector is downstream and unconstrained by this theorem. Status: proven (section 8, depends on E0). Tested by KS-R.2.

Two categories. Everything that moves. Everything that holds.

The entire universe, sorted at birth into two bins by a theorem that has been known since the 19th century.

You already sensed this. Light and gravity. Speed and weight.

The mathematics just proved you right.

4 — The Conjugacy of c and G

Every Planck unit whispers the same secret.

You have never noticed because the textbooks treat it as a dimensional coincidence.

Look closer.

It is not a coincidence. It is a fingerprint.

4.1 — Planck units as evidence

Every Planck unit contains both c and G . Never one alone:

$$l_p = \sqrt{(\hbar G / c^3)} \quad t_p = \sqrt{(\hbar G / c^5)} \quad m_p = \sqrt{(\hbar c / G)}$$

Conventionally treated as dimensional bookkeeping.

AP03 proposes a structural interpretation: c and G always appear together in Planck units because they are not independent.

They are conjugate outputs of the same break.

The Planck scale is the regime where the two modes are barely distinguishable — the scale of the break itself.

4.2 — What is derived and what is forced

In section 7, the minimal pre-geometric action with $\mathbb{Z}_2$ symmetry is constructed.

In the broken phase, perturbations propagate with speed:

$$c^2 = \beta / \alpha \quad (4.1)$$

where α and β here are pre-geometric substrate parameters (temporal and spatial stiffness) — not to be confused with the electromagnetic fine-structure constant α that appears later in this section, which AP06 identifies numerically with the breaking amplitude ε . Context will distinguish them; the substrate parameters appear only inside the action.

A genuine derivation. c emerges as a property of the broken vacuum, not as an input.

Structurally identical to the emergence of the speed of sound in condensed matter.

Theorem E-G — Gravitational Conjugacy. The emergent gravitational coupling G is determined by ε . Originally published as a conjecture; structurally forced by AP20's proof that Axiom B admits exactly one free parameter. The form of G 's dependence on ε , derived in section 7 from the structural condensate density:

$$\mathbf{G} = \mathbf{\kappa} / (\mathbf{\varepsilon} \mathbf{\Lambda}^2) \quad (4.2)$$

Note: the original AP03 ansatz had $c/G \propto \varepsilon^2$, corrected to $c/G \propto \varepsilon$ in the subsequent Sakharov-route derivation. With $G \propto 1/\varepsilon$ at fixed Λ , the Planck mass scales as $m_p \propto \sqrt{\varepsilon}$. This is a substrate-level scaling (κ, Λ free); the parameter-free value of G is derived in AP28 as $G = \alpha_{\text{em}}^{21} (1 + 1/\pi) \hbar c / m_e^2$ (canonical for the value), and the two are reconciled at different architectural levels (see the note on status).

The dimensionless statement of the conjugacy. c and G have different dimensions — c is a velocity, G is a coupling. Their ratio c/G is unit-dependent ($\approx 4.5 \times 10^{18} \approx 10^{19}$ in SI units; trivially $1/G$ in natural units $c = 1$). The unit-independent statement uses the gravitational fine-structure constant at a fixed reference mass:

$$\alpha_{\text{G}}(\mathbf{m}) = \mathbf{G} \mathbf{m}^2 / (\hbar \mathbf{c}) \quad (4.3)$$

With $G = \kappa/(\varepsilon \Lambda^2)$ and the AP06 identification of ε with the electromagnetic fine-structure constant α , the product of the two dimensionless couplings is:

$$\alpha \cdot \alpha_{\text{G}}(\mathbf{m}) = \mathbf{\kappa} \mathbf{m}^2 / (\mathbf{\Lambda}^2 \hbar \mathbf{c}) \quad (4.4)$$

independent of ε (the ε in the numerator from α cancels the $1/\varepsilon$ from G at fixed Λ). This is the dimensionless conjugacy: at any fixed reference mass and cutoff, the product of the electromagnetic and gravitational fine-structure constants is fixed by κ , m , and Λ alone — none of which carry ε -dependence.

The hierarchy, by contrast, scales with ε :

$$\alpha / \alpha_{\text{G}}(\mathbf{m}) \propto \alpha^2 \propto \varepsilon^2 \quad (4.5)$$

and captures the observed hierarchy: at the electron mass, $\alpha/\alpha_G(e) \approx 10^{43}$.

The break controls both. Status: theorem under AP20's forcing proof on Axiom B; the explicit form derived in section 7 from the structural condensate density together with the standard induced-gravity calculation. Tested by KS-R.1 (the conjugacy itself: $\alpha \cdot \alpha_G$ independent of ε) and KS-R.6 (the hierarchy: $\alpha/\alpha_G \propto \varepsilon^2$).

You have just watched a speed limit emerge from nothing.

Not postulated. Not measured. Computed — from the stiffness of the substrate that the break left behind.

The mathematics does not care whether you find this beautiful. It computes.

4.3 — The event horizon as the marginal surface

The Schwarzschild radius of a mass M is:

$$r_s = 2GM / c^2 \quad (4.6)$$

At that radius, the escape velocity equals c .

The propagation mode (Mode 1) can no longer outrun the resistance mode (Mode 0). The break becomes locally marginal.

Inside the horizon, Mode 0 dominates. Outside, Mode 1 has room to run.

Structural interpretation: the event horizon is the surface where the local value of the symmetry-breaking parameter reaches zero.

The boundary between the broken phase (exterior) and the restored phase (interior).

A black hole is a local restoration of the pre-break symmetry.

4.4 — Hawking radiation as proof of global asymmetry

If the interior of a black hole represents local symmetry restoration, then Hawking radiation has a structural interpretation.

The global asymmetry ($\varepsilon > 0$ everywhere in the broken phase) prevents the local restoration from being complete. The break leaks through the horizon.

The global condition $\varepsilon > 0$ is non-negotiable within a universe that exists because $\varepsilon \neq 0$.

A structural reason why black holes must evaporate: a permanently stable local restoration of symmetry within a globally broken phase is a contradiction.

The broken phase is the universe.

The universe does not permit pockets of perfect symmetry to persist indefinitely within itself.

Conjecture E-Hawking. Hawking radiation is the necessary consequence of imposing a global $\varepsilon > 0$ condition on a locally $\varepsilon \rightarrow 0$ region. Status: structurally motivated. Depends on E-Identity, which now inherits from E-G theorem status. Tested via KS-R.4 (horizon inequivalence) — if the cosmological origin and the black-hole horizon are structurally distinct, the structural account of Hawking radiation falls with it.

You have watched a black hole try to undo the break.

And you have watched the universe refuse.

The radiation is the refusal.

The mathematics does not judge. It leaks.

4.5 — The Planck mass as the scale of the break

From $G = \kappa/(\varepsilon\Lambda^2)$ and the definition $m_p = \sqrt{(\hbar c/G)}$:

$$\mathbf{m_p} \propto \sqrt{\varepsilon} \quad (4.7)$$

The Planck mass scales as $\sqrt{\varepsilon}$.

The mass scale at which gravitational and quantum effects are comparable — which is precisely where the two sectors become marginally distinguishable. Tested by KS-R.5.

4.6 — The hierarchy as ε measured

From equation 4.5: $\alpha/\alpha_G(m) \propto \varepsilon^2$ at fixed reference mass m and fixed cutoff Λ .
At the electron mass, $\alpha/\alpha_G(e) \approx 10^{43}$ — read by AP03 as the empirical face of $\varepsilon^2 \approx (1/137)^2 \approx 5.3 \times 10^{-5}$, dressed by the $(\Lambda/m_e)^2$ factor pinned via AP24.

You have wondered why gravity is so absurdly weak compared to electromagnetism.

Forty-three orders of magnitude weaker.

That number is ε , measured.

5 — The Origin and the Horizon

You have looked at the night sky and wondered where it all began.

You have read about black holes and wondered where it all ends.

The chain of reasoning in sections 2 and 4 reveals they are the same place.

5.1 — The same door, two directions

The Big Bang: the moment ε first becomes nonzero. The symmetric state breaks. Mode 1 separates from Mode 0. The universe begins.

Viewed forward in time.

The black hole event horizon: the surface where ε locally returns to zero. Mode 0 recaptures Mode 1. The break closes.

Viewed backward in time.

The same condition — the marginal boundary between broken and unbroken symmetry — encountered from opposite temporal directions.

The Big Bang is the break opening.

The event horizon is the break closing.

The Planck scale is the scale of the break itself.

Claim E-Identity — Origin-Horizon Identity. The cosmological origin and the black hole event horizon are structurally identical — both are the marginal surface $\varepsilon = 0$ of the $\mathbb{Z}_2$ break, traversed in opposite directions. Status: depends on E-G. Since E-G is now structurally forced (Theorem), E-Identity inherits the upgrade. Tested by KS-R.4.

5.2 — The Schwarzschild radius as ratio

A key consequence: the Schwarzschild radius $r_s = 2GM/c^2$ encodes the ratio G/c^2 — Mode 0 over Mode 1 — scaled by mass.

For any given quantity of matter, it identifies where the break becomes marginal again.

The beginning and the end are the same door.

You walk through it in different directions.

You have been walking through it your entire life without knowing.

6 — The Dark Sector Connection

Look at the universe's energy budget.

A pie chart, written by gravitational lensing and the cosmic microwave background.

6.1 — The observed ratio

Dark energy: ~68%. Dark matter: ~27%. Visible matter: ~5%.

The total dark sector constitutes about 95% of the energy budget.

The visible sector — matter that interacts electromagnetically, that propagates and radiates — is the other 5%.

6.2 — Structural expectation

If ε is small, the break displaces only a small fraction of the total content into Mode 1 (the propagation mode).

The majority remains in Mode 0 (the resistance mode).

You already see where this lands.

Ninety-five percent of the universe's energy budget is dark — resistant to electromagnetic interaction, gravitationally dominant.

Qualitatively consistent with a small ε .

A small break means most of what exists stayed behind.

The precise energy partition between Mode 0 and Mode 1 depends on the initial conditions of the symmetry-breaking event, specifically the spectrum of initial fluctuations. The qualitative prediction is robust: for small ε , Mode 0 dominates. The quantitative mapping requires a model of the initial fluctuation spectrum, which remains an open derivation.

6.3 — Caveats

Three issues need resolution before the connection becomes quantitative.

First: dark matter and dark energy are phenomenologically distinct. Dark matter clusters gravitationally; dark energy does not. If both belong to Mode 0, the internal structure of the resistance sector must account for the split.

The pre-geometric action of section 7 structurally provides the split. The action yields two distinct properties for the even sector: the uniform condensate energy density ($\rho_0 = \varepsilon^2 \Lambda^4 / (4\lambda)$, equation 7.9) and the effective mass of localised perturbations ($m^2 = 2\varepsilon \Lambda^2$, equation 7.7). The invariant condensate provides the uniform vacuum baseline — the cosmological-constant-like content that does not cluster, identified phenomenologically as dark energy. The massive localised excitations are the perturbations above the condensate — they have positive mass-energy from the potential curvature, they resist propagation, they cluster gravitationally, identified phenomenologically as dark matter. Both are Mode 0; their phenomenological distinction is the difference between the vacuum baseline and its localised excitations.

This is not a separate hypothesis bolted onto AP03. It is what the action of section 7 already produces. The same equation that gives $c^2 = \beta/\alpha$ and $\rho_0 = \varepsilon^2 \Lambda^4 / (4\lambda)$ also produces $m^2 = 2\varepsilon \Lambda^2$ — the localised-excitation mass scale. The architecture splits Mode 0 into two structural classes at the point where the action is written.

Second: the 68:27:5 ratio is epoch-dependent.

At recombination, the ratio was different. In the far future, dark energy dominates completely. If ε sets the ratio, it must set the initial condition from which the ratio evolves via standard Friedmann cosmology, not the present-day snapshot.

Third: the quantitative mapping $\varepsilon \rightarrow$ (dark/visible ratio) requires a specific functional form, not yet derived.

Conjecture E5 — Dark Sector. The dark-to-visible energy density ratio is determined by ε . Small ε implies most content remains in Mode 0. Status: qualitative on the partition; structural on the internal split (condensate $\rightarrow$ dark energy, localised excitations $\rightarrow$ dark matter). The quantitative functional form is

open work. Tested by KS-R.3. Quantitative follow-up: see AP42 (Predictions Ø Part V).

7 — The Action Principle

You have read the structural argument. Now read where it lives in formal mathematics.

section 7 carries the derivation of c from ε and the structure of G 's determination in the induced-gravity form named in the literature as Sakharov-Visser.

7.1 — Strategy

The simplest possible action for the pre-state S is constructed satisfying three requirements: (i) $\mathbb{Z}_2$ symmetry, (ii) symmetry-breaking vacuum parameterised by a single amplitude ε , and (iii) broken phase yielding two quantities identifiable with c and G , both determined by ε .

The analogue gravity programme (Unruh 1981, Barceló, Liberati & Visser 2005) serves as a structural template, not as a physical identity.

7.2 — The pre-geometric action

Let φ be a real scalar field with $\mathbb{Z}_2$ symmetry ($\varphi \rightarrow -\varphi$). The most general renormalisable potential respecting this symmetry is:

$$V(\varphi) = -\frac{1}{2}\mu^2\varphi^2 + \frac{1}{4}\lambda\varphi^4 \quad (7.1)$$

with $\mu^2 > 0$ (symmetric point unstable) and $\lambda > 0$ (potential bounded below). The minima occur at $\varphi = \pm v$ where $v = \mu / \sqrt{\lambda}$ (7.2). The dimensionless breaking amplitude:

$$\varepsilon \equiv \mu^2 / \Lambda^2 \quad (7.3)$$

where Λ is the fundamental scale. When $\varepsilon \rightarrow 0$, symmetry is restored. When $\varepsilon > 0$, symmetry is broken. The vacuum displacement is $v = \Lambda\sqrt{(\varepsilon/\lambda)}$.

The kinetic structure cannot presuppose a Lorentzian metric (which has not yet emerged). The most general quadratic kinetic term compatible with $\mathbb{Z}_2$:

$$\mathcal{L} = \frac{1}{2}\alpha(\partial_\tau\varphi)^2 - \frac{1}{2}\beta(\nabla\varphi)^2 - V(\varphi) \quad (7.4)$$

where α and β are pre-geometric substrate parameters (temporal and spatial stiffness respectively) and τ is a pre-geometric time parameter. The full action:

$$S[\varphi] = \int d\tau d^3x \left[\frac{1}{2}\alpha(\partial_\tau \varphi)^2 - \frac{1}{2}\beta(\nabla \varphi)^2 - V(\varphi) \right] \quad (7.5)$$

7.3 — The broken phase

The field settles into $\varphi = +v$ (without loss of generality). Write $\varphi = v + \chi$. Expanding to quadratic order:

$$S[\chi] = \int d\tau d^3x \left[\frac{1}{2}\alpha(\partial_\tau \chi)^2 - \frac{1}{2}\beta(\nabla \chi)^2 - \frac{1}{2} m^2 \chi^2 + \dots \right] \quad (7.6)$$

with effective mass:

$$m^2 = V''(v) = 2\mu^2 = 2\varepsilon\Lambda^2 \quad (7.7)$$

The equation of motion yields propagation speed:

$$c^2 = \beta / \alpha \quad (7.8)$$

Result A.1 — Emergence of c. Perturbations on the broken vacuum propagate with speed $c = \sqrt{\beta/\alpha}$. c is not an input but an output of the broken phase. Determined by the ratio of spatial to temporal stiffness of the pre-geometric substrate. Structurally identical to the emergence of the speed of sound in fluid mechanics: $c_s = \sqrt{B/\rho}$. Tested by KS-R.5.

You did not put c into the action.

c came out.

The speed of light is not a law. It is a consequence — of the ratio between how hard the substrate resists deformation in space versus time.

The same mathematics that gives you the speed of sound in water gives you the speed of light in vacuum.

You already knew this intuitively. Sound has a speed because the medium has stiffness. Light has a speed because the vacuum has stiffness.

The universe is a medium. The break made it one.

7.4 — What this action derives

Derived. The propagation speed c (equation 7.8).

Derived. The effective mass of perturbations: $m^2 = 2\varepsilon\Lambda^2$ (equation 7.7). The mass of excitations above the vacuum is set by ε .

Derived. The condensate energy density: $\rho_0 = |V(v)| = \mu^4/(4\lambda) = \varepsilon^2\Lambda^4/(4\lambda)$ (equation 7.9). The energy density of the background condensate — the Mode 0 content. Scales as ε^2 .

7.5 — The form of G 's dependence on ε

The derivation of G was the central open problem at AP03's publication.

AP20's forcing proof on Axiom B shows it could not remain open: Axiom B admits exactly one free parameter, so G cannot be independent of ε . The structural argument is in place before any specific functional form is named. This is the load-bearing step — it comes from the axioms, not from any external scaffolding.

What remains is to find the specific form. The structural inputs are two. First, the condensate energy density from section 7.3:

$$\rho_0 = \varepsilon^2\Lambda^4 / (4\lambda) \quad (7.9 — \text{restated})$$

This is a fact about the broken-phase substrate, derived from the $\{S, B, R, C\}$ -instantiated action. The ε^2 dependence is structural — it follows from the depth of the well $|V(v)| = \frac{1}{4} \mu^4/\lambda$ together with $\mu^2 = \varepsilon\Lambda^2$ (the breaking-amplitude scaling of the potential).

Second, AP20's forcing proof: G has ε -dependence, no other parameter available.

Combining these two structural inputs forces the form. Once the broken vacuum has a finite energy density $\rho_0 \propto \varepsilon^2\Lambda^4$, the induced gravitational coupling on that vacuum has the dimensional form $G \propto 1/(v^2)$ where $v^2 = \mu^2/\lambda = \varepsilon\Lambda^2/\lambda$. Substituting:

$$\mathbf{G} = \mathbf{\kappa} / (\varepsilon\Lambda^2) \quad (7.10)$$

where κ is a dimensionless coefficient set by the structural details of how the broken vacuum responds to perturbation. The Planck mass is $m_p = \sqrt{(\hbar c/G)} \propto \sqrt{\varepsilon}$, modulo the dependence on Λ that AP24 pins.

Correspondence with the literature. The same form appears in standard physics under the name Sakharov-Visser induced gravity — the parallel calculation in standard QFT terms produces exactly this scaling for a scalar-field condensate of our energy density. This is correspondence with our structural result, not its source. Two routes arrive at the same place: the structural route through $\{S, B, R, C\} + \text{AP20} + \text{the section 7.3 condensate density}$, and the literature calculation that produces the same form independently. The literature parallel confirms; it does not derive.

Numerical check. With $\varepsilon \approx 1/137$ (identified with the electromagnetic fine-structure constant via AP06), the dimensionless conjugacy $\alpha \cdot \alpha_G(m) = \kappa m^2/(\Lambda^2 \hbar c)$ is independent of ε ; the hierarchy $\alpha/\alpha_G(e) \approx 10^{43}$ at the electron mass tracks $\varepsilon^2 \cdot (\Lambda/m_e)^2$ once Λ is pinned via AP24. (The dimensionless hierarchy $\alpha/\alpha_G(e) \approx 10^{43}$ is the unit-independent fact; the SI quantity $c/G \approx 10^{19}$ is a different number and must not be conflated with it. AP28 derives this same hierarchy parameter-free as $\alpha_{\text{em}}^{-20}/(1 + 1/\pi) \approx 4 \times 10^{42}$, dissolving it as a channel count rather than a fine-tuning.)

Status. The ε -dependence of G is theorem under AP20's forcing proof. The specific functional form is derived in this section from the structural condensate density. The structural coefficient κ is not derived from $\{S, B, R, C\}$ alone; computing it requires the response calculation of the broken vacuum, which leans on the standard QFT cutoff structure of the literature. This is debt D1, partially closed by AP08 + this paper, not fully closed (see section 0.6).

7.6 — Summary

Established. (1) The $\mathbb{Z}_2$ -symmetric double-well action is the minimal pre-geometric theory with SSB. (2) In the broken phase, perturbations propagate with speed $c = \sqrt{(\beta/\alpha)}$. (3) The condensate energy density scales as ε^2 . (4) Via AP20 forcing + the section 7 derivation: $G = \kappa/(\varepsilon\Lambda^2)$, $c/G \propto \varepsilon$, $m_p \propto \sqrt{\varepsilon}$.

Open. (1) The substrate parameters α and β are introduced but not derived; c is parameterised but not predicted (debt D-AP03.1). (2) The kinetic term presupposes a distinction between temporal and spatial coordinates (τ vs x) — a structural input, not an output. The deepest open problem in AP03, distinct

from and prior to the G derivation. A truly pre-geometric formulation would derive this distinction.

8 — The $\mathbb{Z}_2$ Representation Theorem

You have followed the structural argument that the universe sorts into two sectors at birth.

This section carries the mathematics behind that claim.

8.1 — Statement

Theorem B.1. The group $\mathbb{Z}_2 = \{e, g\}$ where $g^2 = e$ has exactly two irreducible representations over the real (or complex) numbers: the trivial representation and the sign representation.

8.2 — Proof

Let $\rho: \mathbb{Z}_2 \rightarrow \text{GL}(V)$ be a representation of $\mathbb{Z}_2$ on a vector space V . The generator g satisfies $g^2 = e$, so $\rho(g)^2 = I$ (the identity matrix).

Therefore $\rho(g)$ is an involution: its eigenvalues satisfy $\lambda^2 = 1$, giving $\lambda = +1$ or $\lambda = -1$.

The vector space V decomposes into eigenspaces:

$$V = V_+ \oplus V_- \quad (8.1)$$

where V_+ is the $+1$ eigenspace ($\rho(g)v = +v$ for all $v \in V_+$; vectors invariant under the $\mathbb{Z}_2$ action; the trivial representation), and V_- is the -1 eigenspace ($\rho(g)v = -v$ for all $v \in V_-$; vectors that flip sign under the $\mathbb{Z}_2$ action; the sign representation).

Each eigenspace is itself an invariant subspace and carries an irreducible representation of $\mathbb{Z}_2$. Since the only eigenvalues are $+1$ and -1 , there are exactly two irreducible representations.

Any representation of $\mathbb{Z}_2$ decomposes into copies of these two.

There is no third option.

The mathematics does not negotiate.

Two irreducible representations. Two sectors. Two speed limits. Two fundamental categories.

The universe at birth: sorted into two bins by a theorem proved before electromagnetism was a theory.

9 — Kill Switches

AP03 is falsified if any of the following are demonstrated. Each kill switch targets a specific claim. If any claim falls, the dependent claims fall with it, but independent claims survive.

Here is how to break it. The argument does not hide. It shows you exactly where to strike.

KS-R.1 — Independent variability of c and G

Claim. The two fundamental speed limits c and G are not independent. Both are determined by the single breaking amplitude ε . Originally published as conjecture E-G; upgraded to a theorem by AP20's forcing proof that Axiom B admits exactly one free parameter.

Test. Demonstrate that the dimensionless gravitational coupling $\alpha_G(m)$ at a fixed reference mass varies independently of the electromagnetic fine-structure constant α — that α_G and α can be tuned independently with no common ancestor parameter. Quasar absorption spectra constrain cosmological variation of α ; lunar laser ranging and Big Bang nucleosynthesis constrain α_G . AP03 predicts these variations are correlated. Uncorrelated variation falsifies the conjugacy.

Status. LIVE — EMPIRICAL.

Recovery. If KS-R.1 fires, both the conjugacy claim and AP20's forcing proof on Axiom B are in conflict with measurement. Either the forcing proof is wrong (Axiom B admits more than one free parameter) or AP03's identification of the parameter with ε is wrong (the parameter is something else). The 420 Code retreats by reopening Axiom B's formal status and acknowledging that the constants are independent within the architecture as currently stated. Section 4 collapses; sections 5 and 6 inherit the collapse.

KS-R.2 — Non-binary origin

Claim. The symmetry-breaking event produces exactly two fundamental sectors because the symmetry group of the pre-state is $\mathbb{Z}_2$, whose representation theory

admits exactly two irreducible representations. Mode 1 carries propagation (c); Mode 0 carries resistance (G). Claim E2.

Test. Demonstrate that the early universe requires more than two fundamental mode sectors at the moment of symmetry breaking — not merely more than two particle species, but more than two irreducible sectors of the symmetry decomposition. Detection of a third fundamental sector that is neither gravitationally coupled (Mode 0) nor electromagnetically coupled (Mode 1) triggers this kill switch.

Status. LIVE — EMPIRICAL.

Recovery. If KS-R.2 fires, the $\mathbb{Z}_2$ postulate (E0) fails and the pre-state must have a richer symmetry. The structural claim that exactly two speed limits emerge collapses. The 420 Code retreats by upgrading the pre-state symmetry to whatever the data forces — $\mathbb{Z}_3$, $U(1)$, $SU(2)$, or richer — and re-deriving the sector content under the new representation theory. Sections 3 and 8 are reworked; section 4 retains the structure but expands to N sectors.

KS-R.3 — Dark ratio independence

Claim. The dark-to-visible energy density ratio ($\sim 95:5$) is determined by ε . Small ε implies most content remains in Mode 0. Conjecture E5.

Test. Demonstrate that the dark-to-visible energy density ratio is set by a mechanism entirely independent of the c/G relationship and of the breaking amplitude ε — for example, by a hidden gauge sector with its own coupling structure unrelated to the $\mathbb{Z}_2$ break.

Status. LIVE — EMPIRICAL.

Recovery. If KS-R.3 fires, the qualitative link between ε and the dark-sector partition collapses. Conjecture E5 is falsified. The 420 Code retreats by treating the dark-sector partition as a separate open problem independent of AP03, possibly involving additional fields or sectors not captured by the minimal $\mathbb{Z}_2$ architecture. Section 6 is removed; sections 1–5 and 7–8 are unaffected.

KS-R.4 — Horizon inequivalence

Claim. The cosmological origin and the black hole event horizon are structurally identical — both are the marginal surface $\varepsilon = 0$ of the $\mathbb{Z}_2$ break, traversed in opposite temporal directions. Claim E-Identity. Originally LIVE — HARD; inherits the structural-forcing upgrade from KS-R.1.

Test. Demonstrate that the singularity theorems have fundamentally different character for cosmological versus black-hole singularities in a way incompatible with the $\varepsilon = 0$ interpretation — for example, by showing that the cosmological singularity has a structural property (handedness, dimension, residual entropy) that the black hole singularity provably cannot have within the AP03 architecture.

Status. LIVE — HARD.

Recovery. If KS-R.4 fires, the structural identification of the two surfaces fails. Section 5 collapses. The 420 Code retreats to the position that cosmological and black-hole boundaries are distinct structural surfaces — each marginal in its own way but not the same surface. AP03's claim that the beginning and the end are the same door is withdrawn. Sections 1–4, 6–8 are unaffected.

KS-R.5 — Planck scale mismatch

Claim. The propagation speed c emerges from the broken vacuum as $c = \sqrt{(\beta/\alpha)}$ (Result A.1, equation 7.8), and the Planck mass scales as $m_p \propto \sqrt{\varepsilon}$ (equation 4.7). The Planck scale is the scale of the break itself.

Test. Demonstrate that the propagation speed of perturbations on the broken vacuum is parametrically different from $\sqrt{(\beta/\alpha)}$, or that the Planck-scale onset is shown incompatible with $m_p \propto \sqrt{\varepsilon}$. The empirical handles are precision tests of Lorentz invariance at high energy (deviation from c at the Planck scale) and high-energy cosmic-ray observations of the effective UV cutoff.

Status. LIVE — EMPIRICAL.

Recovery. If KS-R.5 fires, the action principle of section 7 fails. c is not derived as $\sqrt{(\beta/\alpha)}$; the kinetic term needs revision, or the broken-phase analysis is incomplete. Result A.1 collapses; the conjugacy claims that depend on it weaken to qualitative form. Sections 1–3, 5–6, 8 stand.

KS-R.6 — Dimensionless scaling violation

Claim. The dimensionless conjugacy holds: at fixed reference mass m and fixed cutoff Λ , the product $\alpha \cdot \alpha_G(m)$ is independent of ε , and the ratio $\alpha/\alpha_G(m)$ scales as ε^2 (equations 4.4 and 4.5). The dimensionful gravitational hierarchy $ke^2/(G \cdot m_e) \approx 10^{42}-10^{43}$ is the SI illustration of the same fact; the unit-independent statement is the dimensionless one. Originally registered with the dimensionful form $c/G \propto \varepsilon^2$ in the AP03 conjecture; the dimensionless statement is the corrected scaling actually produced by the derivation.

Test. Demonstrate that the product $\alpha \cdot \alpha_G(m)$ varies measurably with ε — either across cosmological epochs (where α may vary, per the AP06 identification of ε with α) or at fixed-mass laboratory comparisons — or that the hierarchy $\alpha/\alpha_G(\text{electron})$ is not consistent with ε^2 scaling once the $(\Lambda/m_e)^2$ factor is fixed via AP24. The empirical handles are precision tests of cosmological variation of the electromagnetic fine-structure constant (quasar absorption spectra) and laboratory tests of the gravitational fine-structure constant at fixed reference masses (lunar laser ranging, Big Bang nucleosynthesis). Scope: the ε^2 scaling tested here is the substrate-level statement at fixed κ and Λ (κ, Λ free, debt D-AP03.1); the physical response of the hierarchy to a measured variation of α is carried by AP28's canonical form ($G \propto \varepsilon^{21}$, hierarchy $\propto \varepsilon^{-20}$) and is tested there.

Status. LIVE — HARD. Superseded as the statement of G 's value by AP28 (registry status: SUPERSEDED AS VALUE — see AP28); the switch's live content is the substrate-level statement at fixed κ and Λ .

Recovery. If KS-R.6 fires, the form of G 's dependence on ε derived in section 7 is wrong. The 420 Code retreats by treating G 's specific dependence on ε as governed by a more complex relationship — possibly involving additional structural quantities beyond ε . Section 4.2 and section 7.5 are reworked; section 4.5 ($m_p \propto \sqrt{\varepsilon}$) inherits the change. Section 5 may survive in modified form; section 6 unaffected.

Why these six and not more

The six kill switches above cover the load-bearing claims of the paper as registered in the master kill switch register. Each closure would force a different kind of retreat.

KS-R.1 closes the conjugacy of c and G — the central structural claim of the paper. KS-R.2 closes the binary decomposition by undermining $\mathbb{Z}_2$. KS-R.3 closes the dark-sector connection. KS-R.4 closes the origin-horizon identity. KS-R.5 closes the action-principle derivation of c . KS-R.6 closes the specific dimensionless scaling.

Two structural claims that some readings would expect to carry their own formal kill switch — the $\mathbb{Z}_2$ minimality postulate (E0) and the Hawking-radiation structural interpretation (E-Hawking) — are absorbed into the kill switches above. E0's minimality is tested by KS-R.2: any falsification of the binary decomposition (more than two fundamental sectors) is also a falsification of $\mathbb{Z}_2$ as the minimal symmetry, since a richer pre-state symmetry would produce more than two irreducible representations. E-Hawking's structural interpretation is tested by KS-R.4: the origin-horizon identity is what makes Hawking radiation a structural consequence of global asymmetry; if KS-R.4 fires (cosmological origin and black-hole horizon are not the same marginal surface), the structural account of Hawking radiation falls with it. Both claims are thus held to formal kill-switch standard without adding entries to the master register.

Smaller claims throughout the paper — the specific values of $\alpha_G(e)$, the Schwarzschild interpretation as Mode 0 dominance — are illustrative rather than load-bearing. If they turn out to be poor illustrations, the paper still stands. If KS-R.1 through KS-R.6 hold, the structural argument of AP03 holds.

10 — What This Establishes and What Remains Open

10.1 — What this paper closes

The conjugacy of the two fundamental speed limits c and G , derived from $\{S, B, R, C\}$ via AP03's postulates E0 and E1.

Specifically:

First. The bootstrap problem solved. At the moment of symmetry breaking, only the breaking amplitude ε is available as a scale. Both c and G must therefore inherit from ε . The argument is structural, forced by the absence of any other available quantity at the relevant moment.

Second. The binary decomposition proven. The $\mathbb{Z}_2$ representation theorem (section 8) decomposes the broken phase into exactly two irreducible sectors. Mode 1 carries propagation (c); Mode 0 carries resistance (G). No third fundamental sector is possible within the $\mathbb{Z}_2$ architecture.

Third. The propagation speed c derived from a minimal pre-geometric action. $c = \sqrt{(\beta/\alpha)}$, where β and α are the spatial and temporal stiffness of the pre-geometric substrate. c is not an input but an output of the broken phase, structurally identical in form to the emergence of the speed of sound in fluid mechanics.

Fourth. The gravitational coupling G structurally forced to depend on ε . AP03 alone presented this as conjecture E-G; AP20's forcing proof on Axiom B upgrades it to a theorem. The explicit scaling $G = \kappa/(\varepsilon\Lambda^2)$, equivalently $c/G \propto \varepsilon$, was derived from the structural condensate density of section 7 in subsequent work and verified numerically (the gravitational hierarchy $ke^2/(G \cdot m_e^2) \approx 10^{42}-10^{43}$ at $\varepsilon \approx 1/137$).

Fifth. The structural identity of the cosmological origin and the black hole event horizon. Both are the marginal surface $\varepsilon = 0$ of the $\mathbb{Z}_2$ break, traversed in opposite temporal directions. The Big Bang is the break opening; the event horizon is the break closing. The beginning and the end are the same door.

Sixth. The structural account of Hawking radiation as the universe refusing to permit a permanent pocket of restored symmetry within a globally broken phase. Conjecture E-Hawking; structurally motivated, empirically tested.

Seventh. The qualitative connection between small ε and the observed dark-to-visible energy ratio. Conjecture E5; qualitatively consistent with the 95:5 observation but not quantitatively derived.

10.2 — What this paper does not close

Two structural debts remain open and are named explicitly. These are unresolved derivations tied to specific structural claims.

D1 (registry-level, partially closed). The first-principles derivation of G from the axioms without external scaffolding. AP03 contributes by establishing that G cannot be independent of ε (via AP20's forcing proof) and by deriving the explicit form $G = \kappa/(\varepsilon\Lambda^2)$ from the structural condensate density of section 7. The structural coefficient κ is not derived from $\{S, B, R, C\}$ alone; computing it leans on standard QFT cutoff machinery. The closure is partial. Full statement at section 0.6.

D-AP03.1. The pre-geometric substrate parameters α and β are introduced but not derived from a more fundamental theory. Their ratio gives $c^2 = \beta/\alpha$, but neither α nor β individually is computed from the axioms. The kinetic term $\mathcal{L} = \frac{1}{2}\alpha(\partial_\tau\varphi)^2 - \frac{1}{2}\beta(\nabla\varphi)^2 - V(\varphi)$ presupposes a distinction between temporal and spatial coordinates (τ vs x) — a structural input, not an output. The deepest open problem in AP03 — distinct from and prior to the G derivation. A truly pre-geometric formulation would derive this distinction from the axioms rather than presupposing it.

10.3 — Items held open without claiming debt status

Two further items are bracketed inside this paper. They are real and live, but they are not structural debts in the same sense as D1 and D-AP03.1.

E5 quantitative. The functional form linking ε to the dark-to-visible energy ratio. The paper establishes the qualitative direction (small $\varepsilon \rightarrow$ most content in Mode 0) but does not derive the quantitative mapping. Three issues need resolution before the connection becomes quantitative: the phenomenological

distinction between dark matter and dark energy within Mode 0; the epoch-dependence of the 95:5 ratio over cosmological history; the specific functional form linking ε to the partition. This is open derivation work for Notebook VI; the qualitative claim stands.

E-Hawking sufficiency. Hawking radiation is structurally motivated in section 4.4 as the necessary consequence of imposing a global $\varepsilon > 0$ condition on a locally $\varepsilon \rightarrow 0$ region. The paper does not derive the Hawking spectrum from the AP03 action; it argues that the global condition forces leakage and identifies the spectrum derived elsewhere with that leakage. The empirical test rides on KS-R.4 (horizon inequivalence): if the cosmological origin and the black-hole horizon are shown structurally distinct, the structural account of Hawking radiation falls with it. Open work: deriving the spectrum directly from the AP03 architecture.

10.4 — Where the philosophical-register implementations live

AP03 is a physics derivation. Its philosophical-register companions — the chapters where the same structural facts are inhabited in the prose register of the $\emptyset$ Models catalogue — are named here at the level of expected pairings; final placement is confirmed during the $\emptyset$ Models compilation sweep alongside the Notebook II audit programme.

Predictions $\emptyset$ (Part V — dark-sector partition and Hubble structure) carries the empirical face of AP03's conjecture E5: the predictions catalogue treats the 5/27/68 visible-dark structure (visible matter $\sim 5\%$, dark matter $\sim 27\%$, dark energy $\sim 68\%$, matching Planck 2018) as a prediction grounded in the small ε of the break, with the testable numerical handles. Predictions $\emptyset$ inherits AP03's qualitative direction (small $\varepsilon \rightarrow$ Mode 0 dominance) and tightens it where a numerical form is available downstream of AP06 and AP24.

Resolutions $\emptyset$ (Chapter 4 — Laws of Nature) carries the philosophical companion to AP03's structural claim that c and G are not independent constants. The Resolutions chapter treats laws of nature as what the four conditions $\{S, B, R, C\}$ structurally produce when read at a specific resolution — c and G are two of those laws at the spacetime resolution, both inheriting from the same break. AP03 provides the formal derivation; Resolutions Ch 4 provides the philosophical register in which The 420 Code says it.

The direction of acknowledgment is two-way. AP03 is the original derivation — the conjugacy of c and G is established here in formal mathematics. The Ø Models chapters extend, inhabit, and translate the result into the prose register of the catalogue. The 420 Code reads AP03 forward into the Ø Models; the Ø Models read back into AP03 by carrying its conclusions into questions the formal paper does not directly address. Final companion-pairing list is fixed at Ø Models compilation.

10.5 — Structural relationships to subsequent work

AP05 (The Break — Axiom $\rightarrow$ Structure) follows AP03 in Notebook II. AP05 walks the seven-step chain from $1:1 + 1 \times \varepsilon$ @ AS to obligation and reformulates the cosmological identity as Israel junction conditions on the completion boundary Σ — a structural alternative to dynamical bounce cosmologies. AP05 inherits AP03's broken phase as the substrate on which the metric is built. Where AP03 derives the conjugacy of c and G , AP05 derives the metric structure itself.

AP10 (The Dimension) follows AP05 in Notebook II. AP10 derives $N = 3$ spatial dimensions from the four axioms. Combined with AP08, AP10 closes the central conditional in AP08's derivation of Einstein's field equations. AP10 sits structurally downstream of AP03 because the binary decomposition (Mode 0 / Mode 1) sets up the manifold on which the dimensional count operates.

AP08 (The Identity) closes Notebook II. AP08 derives Einstein's field equations from the record algebra. The gravitational coupling G that appears in AP08's field equations is the same G that AP03 conjectures (and AP20 forces) to be determined by ε . AP08's disclosed circularity in $G = 2\kappa/m_e^2$ is the registry-level structural debt D1, substantially reduced by the AP20 forcing proof + the section 7 derivation from AP03 but not fully closed.

AP06 (The Leakage Constant, Notebook IV) identifies ε numerically with the fine-structure constant $\alpha \approx 1/137$. AP06's cycle form $1:1 + 1 \times \varepsilon$ @ AS [$+1/137$ / $-1/137$] is what makes AP03's $c/G \propto \varepsilon$ numerically verifiable: the gravitational hierarchy $ke^2/(G \cdot m_e^2) \approx 10^{42}-10^{43}$ at $\varepsilon \approx 1/137$ matches observation.

AP24 (The Residual, Notebook IV) constrains the cutoff scale Λ that pins AP03's scaling relations to specific numbers. With AP24 in place, $m_p \propto \sqrt{\varepsilon}$ becomes $m_p \propto \sqrt{\varepsilon} \cdot \sqrt{\Lambda}$ with a specific numerical Λ .

AP17 (The Room, Notebook VI) relates the 0-pole / 1-pole structure of cosmological geometry to AP03's Mode 0 / Mode 1 sectors. AP17 inherits AP03's binary decomposition as the structural skeleton of cosmological observations.

AP18 (Notebook VI) derives the MOND floor $a_0 \approx cH_0$ from the axioms, downstream of the small ε . AP18 reads AP03's result as constraining the structural form the MOND-scale acceleration must take.

10.6 — Placement within The 420 Code

AP03 opens Notebook II (Spacetime). The chapter sequence runs AP03 — AP05 — AP10 — AP08. AP03 sits first because it does the constants: c and G are the structural ground on which AP05's Lorentzian metric, AP10's dimensional count, and AP08's field equations all stand.

Within the reading order of The 420 Code, AP03 follows the three chapters of Notebook I (AP40 The Irrational, AP01 The Actualization State, AP20 The Proof). Notebook I establishes the axiom, names the four conditions, and proves the Embedding Hypothesis. Notebook II runs unconditionally on what Notebook I closes.

AP03 is logically independent of AP01 and AP40 in the strict sense: its derivations require only standard representation theory and scalar field theory, and they do not invoke EH, QRA, or the Actualization State apparatus. What AP03 does invoke from Notebook I is the axiomatic floor $\{S, B, R, C\}$ and AP20's forcing proof on Axiom B (which upgrades E-G from conjecture to theorem). Without AP20, AP03 still derives c and conjectures the conjugacy of c and G . With AP20, the conjugacy is structurally forced.

10.7 — How The 420 Code reads now

AP03 in place changes The 420 Code reading in three ways.

First. The fundamental constants are no longer treated as independent inputs of physics. c and G are conjugate outputs of the same break — both determined by ε . The 420 Code says this plainly. Future chapters do not need to re-derive it; they inherit it.

Second. The cosmological origin and the black hole event horizon are read as the same structural surface, traversed in opposite directions. Notebook VI (Cosmology) inherits this reading. The cosmology chapter does not need to separately justify treating the Big Bang singularity and the horizon as structurally related; AP03 has done it.

Third. The dark-sector partition is read as a consequence of small ε . The qualitative direction is in place. Notebook VI takes up the quantitative work — deriving the specific functional form linking ε to the 95:5 ratio.

These three structural facts — the conjugacy, the origin-horizon identity, the dark-sector qualitative connection — are what AP03 contributes to The 420 Code reading. They are inherited by every subsequent chapter that touches spacetime, the constants, or the cosmological structure.

10.8 — The closing

You have walked through this paper. You have seen the bootstrap problem named, the binary decomposition proven, the propagation speed c derived from a minimal pre-geometric action, the gravitational coupling G structurally forced to depend on ε , the cosmological origin and the black hole event horizon identified as the same marginal surface, the dark sector qualitatively connected to the smallness of ε .

What follows in your reading life is what follows in any reader's life when a structural fact has been seen clearly. The constants you grew up reading as independent are not independent. The Big Bang and the black hole are not two events but two sides of one boundary. The dark sector is not anomaly but expected consequence.

The break controls everything.

One amplitude. ε .

From it: the speed at which light crosses the room before your hand finishes reaching for the switch, and the weight that holds your foot to the floor, and the boundary the universe began at, and the boundary every black hole repeats. All from the same source.

The mathematics does not negotiate. It computes. It computes the same answer twice — once for c , once for G — and the answer is ε .

Two speed limits.

One break.

The same door, walked from both sides.

Claim Summary

section 0 (Dependency and Scope). Contract with the reader. Dependencies, axiom mapping, epistemic status per section, kill switch summary, debts owed.

section 1 (The Two Speed Limits). FRAMING. The two structural speed limits of physics — c (propagation, universal) and G (the gravitational rate bound, system-dependent) — named as siblings. Both non-negotiable, both structural, governing different processes.

section 2 (Before the Arena). STRUCTURAL ARGUMENT. The bootstrap problem: at the moment of symmetry breaking, no spacetime, no mass, no pre-existing coupling. The only available scale is the breaking amplitude itself. Claim E-Bootstrap: ε is the first speed limit; both c and G inherit from it.

section 3 (The Minimal Break). DERIVATION. The pre-state has $\mathbb{Z}_2$ symmetry (Postulate E0). The $\mathbb{Z}_2$ representation theorem (Theorem 3.1; proof in section 8; tested by KS-R.2) decomposes the broken phase into exactly two irreducible sectors. Mode 1 (odd sector) carries propagation; Mode 0 (even sector) carries resistance. Claim E2.

section 4 (The Conjugacy of c and G). DERIVATION + STRUCTURAL CRITIQUE. Planck units as evidence (every Planck unit contains both c and G). c derived from the pre-geometric action: $c^2 = \beta/\alpha$ (Result A.1, section 7). Theorem E-G: $G = \kappa/(\varepsilon\Lambda^2)$, with the dimensionless conjugacy $\alpha \cdot \alpha_G(m)$ independent of ε and the hierarchy $\alpha/\alpha_G \propto \alpha^2$, structurally forced by AP20 (tested by KS-R.1 and KS-R.6). The event horizon as the marginal surface $\varepsilon = 0$. Hawking radiation as proof of global asymmetry (Conjecture E-Hawking, tested via KS-R.4). The Planck mass scales as $\sqrt{\varepsilon}$ (tested by KS-R.5).

section 5 (The Origin and the Horizon). DERIVATION. Claim E-Identity: the cosmological origin and the black hole event horizon are structurally identical — both the marginal surface $\varepsilon = 0$, traversed in opposite temporal directions. The Big Bang opens the break; the event horizon closes it. The Schwarzschild radius encodes the ratio G/c^2 — Mode 0 over Mode 1. Tested by KS-R.4.

section 6 (The Dark Sector Connection). QUALITATIVE FRAMING. The observed dark-to-visible energy ratio (95:5). Structural expectation: small ε implies most content remains in Mode 0. Three caveats: dark matter / dark

energy distinction within Mode 0; epoch-dependence of the ratio; functional form not yet derived. Conjecture E5; tested by KS-R.3.

section 7 (The Action Principle). FORMAL DERIVATION. Strategy: $\mathbb{Z}_2$ -symmetric pre-geometric action with single breaking amplitude. The action $S[\varphi] = \int d\tau d^3x [1/2\alpha(\partial_\tau\varphi)^2 - 1/2\beta(\nabla\varphi)^2 - V(\varphi)]$ with $V(\varphi) = -1/2\mu^2\varphi^2 + 1/4\lambda\varphi^4$. Broken-phase expansion yields $c^2 = \beta/\alpha$ (Result A.1) and effective mass $m^2 = 2\varepsilon\Lambda^2$. Condensate energy density $\rho_0 = \varepsilon^2\Lambda^4/(4\lambda)$. Form of G 's dependence on ε derived from the condensate density: $G = \kappa/(\varepsilon\Lambda^2)$, with the dimensionless conjugacy $\alpha \cdot \alpha_G(m)$ independent of ε (numerical check: the gravitational hierarchy $\kappa e^2/(G \cdot m_e^2) \approx 10^{42}$ – 10^{43} in SI is the unit-dependent face at $\varepsilon \approx 1/137$). The induced-gravity form is named in the literature as Sakharov-Visser; the literature calculation is correspondence, not source. D-AP03.1: α and β introduced but not derived; deepest open problem.

section 8 (The $\mathbb{Z}_2$ Representation Theorem). FORMAL PROOF. Theorem B.1: $\mathbb{Z}_2$ has exactly two irreducible representations — trivial and sign. Proof: the generator g satisfies $g^2 = e$, so $\rho(g)^2 = I$; eigenvalues ± 1 ; V decomposes into $V_+ \oplus V_-$. No third option.

section 9 (Kill Switches). FORMAL FALSIFICATION REGISTER. KS-R.1 (independent variability of c and G , LIVE — EMPIRICAL); KS-R.2 (non-binary origin, LIVE — EMPIRICAL); KS-R.3 (dark ratio independence, LIVE — EMPIRICAL); KS-R.4 (horizon inequivalence, LIVE — HARD; absorbs the E-Hawking structural test); KS-R.5 (Planck scale mismatch, LIVE — EMPIRICAL); KS-R.6 (dimensionless scaling violation, LIVE — HARD).

section 10 (What This Establishes and What Remains Open). SYNTHESIS. Seven structural closures named (bootstrap, binary decomposition, c derived, G 's ε -dependence forced by AP20 with the explicit form derived in section 7, origin-horizon identity, Hawking structural account, dark sector qualitative). Two structural debts (D1 partially closed; D-AP03.1 substrate parameters). Two items held open without claiming debt status (E5 quantitative; E-Hawking sufficiency). Placement within The 420 Code: opens Notebook II (Spacetime). Closing on the break controlling everything.

Conditionality Footer

Dependencies

AP20 (the canonical axioms {S, B, R, C}; formal statement section 3.3; obligations fenced at KS-P.1/KS-P.2).

AP20 (forcing proof on Axiom B — upgrades E-G from conjecture to theorem).

AP06 (Leakage Constant — numerical identification $\varepsilon \approx 1/137$).

AP24 (Residual — cutoff scale Λ that pins m_p scaling).

Dependents

AP05 (The Break) inherits AP03's broken phase as the substrate on which the Lorentzian metric is built.

AP08 (The Identity) uses the same G that AP03 conjectures (AP20 forces) to be determined by ε ; AP03's section 7 derivation of G's functional form from the structural condensate density partially closes the registry-level debt D1 disclosed in AP08's Lock.

AP17 (The Room) inherits the binary decomposition Mode 0 / Mode 1 as the structural skeleton of cosmological geometry.

AP18 (MOND) reads AP03's small ε as constraining the structural form of the MOND-scale acceleration.

Predictions $\emptyset$ (Part V) inherits AP03's qualitative dark-sector direction (small $\varepsilon \rightarrow$ Mode 0 dominance) as a testable prediction.

Resolutions $\emptyset$ (Chapter 4) inherits AP03's structural conjugacy claim as part of the philosophical-register account of laws of nature.

Kill switches closed by this paper

None explicitly closed. AP03 opens six kill switches (KS-R.1 through KS-R.6) and contributes the AP20-forced upgrade of E-G from conjecture to theorem.

The paper does retroactively strengthen the account in The 420 Code of the fundamental constants — c and G are no longer treated as independent inputs but as conjugate outputs of the same break.

Kill switches that remain live

KS-R.1 through KS-R.6. See section 9 for the full register.

Structural debts owed

D1 (registry-level, partially closed). The first-principles derivation of G from the axioms without external scaffolding. AP03 reduces D1 substantially via AP20's forcing proof plus the section 7 derivation of G 's functional form from the structural condensate density, but does not fully close it. Full statement at section 0.6.

D-AP03.1. The pre-geometric substrate parameters α and β introduced but not derived; the kinetic term presupposes the distinction between temporal and spatial coordinates. Deepest open problem in AP03.

Items held open without claiming debt status

E5 quantitative. The functional form linking ε to the dark-to-visible energy ratio. Qualitative direction in place; quantitative mapping open for Notebook VI.

E-Hawking sufficiency. Direct derivation of the Hawking spectrum from the AP03 architecture (structural motivation provided in section 4.4; full derivation open).

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